



TLD Group Engineering Guidelines

The TLD Group is a decentralized organisation where each Industrial entity is fully responsible for the development of the products it has been given the responsibility to engineer and to manufacture. The key local people accountable for the product development are the Engineering Manager and the Chief Operation Officer of the industrial BU.

However, some coordination and controls are necessary at the TLD group level for the following reasons:

- o The customer sees and must see TLD Group as a unique and consistent entity
- o Similar products can be manufactured in several TLD industrial sites
- o Part supports and Cost optimization require standardization above the factory level
- o Introduction of any new technology must be done after we insure that it is reliable and the TLD service organization and customers could handle it in the fields

The TLD Group Technical Director (GTD) is in charge of the coordination of all engineering activities of the TLD group.

The TLD Group Technical Director works directly with the Industrial COO's and with the EM's.

In parallel of validating and controlling some engineering development and proposals engineering group, the GTD role is also to promote innovation, standardization and benchmarking between different TLD factories and insure the development of common processes.

Engineering activity:

- A. **Product drawing correction or minor adaptation for better produce-ability:**
- B. **Product on going improvement and customization of a product** (Ex: new handrail for a NBL, design of customer specific options ...)
 - a. As long as it does not involve new technologies or it does not affect the TLD Standardization or RANGER products, the engineering development is managed by the EM under the supervision of the COO.
 - b. If new technologies are to be used, the TLD Group Technical director should be consulted and the new system should be considered as a Type D project.

- c. If it affects the TLD standardization or the RANGER products, the TLD Group technical director and/or the other EM's should be consulted before engaging the improvement and the final decision should be approved by all involved parties.

C. **New Product Development or complete upgrade of a TLD Product line**
(Ex: LSP development, redesign of the TF-7-GR)

At TLD, any new product development launch (start of PACE Phase 0) needs to be pre-approved by the TLD Steering committee after submission to the TLD Group COO.

The next step of the development (PACE Phase 0-> phase 1) will also have to be approved by the Steering committee and at this stage, a complete TLD New Product development package will have to be filled out (control by the GTD before submission) and will have to contain the information requested to exit Phase 0 (see below chart).

Once the phase 0 of the project has been approved, the project will have to continue to be managed and controlled according to the PACE mode and with the relevant approvals at each phase (table here below).

D. **Significant improvement/evolution/technologic change of a TLD Product line**
(Ex: Engine evolution, new cab evolution, control system change)

Any significant improvement/evolution at TLD needs to be pre-approved by the GTD and then the TLD Group COO.

Once the start of phase 0 of the project has been approved, the project will have to continue to be managed and controlled according to the PACE mode and with the relevant approvals at each phase (table here below).

PACE (Product And Cycle-time Excellence) process:

PACE Phase transition	PACE Phase	Phase Description	Mandatory formal approvals for phase transition			
			Dev type #A	Dev type #B	Dev type #C	Dev type #D
Launch of Phase 0			Engineer	EM	TLD Steering ctee	GTD & TLD Group COO
	PHASE 0	Pre-project				
Phase 0 ->1			N/A	N/A	TLD Steering ctee	EM, COO, CEO, GTD
	PHASE 1	Project				
Phase 1->2			EM	EM	EM, COO, CEO, GTD and Group COO	EM, COO, CEO, GTD
	PHASE 2	Engineering				
Phase 2->3			EM	EM & COO	EM, COO, CEO, GTD and Group COO	EM, COO, CEO, GTD
	PHASE 3	Prototype				
Phase 3->4			N/A	N/A	EM, COO, CEO, GTD and Group COO	EM, COO, CEO
	PHASE 4	Industrialisation				
End Project			N/A	N/A	COO & CEO	COO & CEO

All validations phase highlighted in Yellow should be done through the EAP module and according to the EAP rules (see EAP help page).

Phase validation may require a formal meeting. Some project may require some intermediate review “critical design review”.

The minimum typical criteria to exit one phase for project B & C are given in the below chart.

Minimum Criteria to fulfil to Exit each phase				
Phase 0>1	Phase 1>2	Phase 2>3	Phase 3>4	Exit Phase 4
Product description and features Volumes expected	Updated Project Schedule Basic layout and concept drawings	Updated Project Schedule Drawings set: Structure, weldments, assembly drawing	Updated Project Schedule Major components validations: Engine approval,...	BOM and Packages review Data base (Doc convergence)
Estimated Market	Preliminary schematics and flow diagrams	Finalized flow, electric, and hydraulic drawings	Test reports	Price list
Projected volumes for TLD	Refine spec and performance requirement	List of deviation to the validated spec and performances	List of deviation to the validated spec and performances	Final List of deviation to the validated spec and performances
Development cost		BOM structure	Internal review report	Development cost review
DTC (Design To Cost) documentation Target cost for the unit	Updated DTC	Updated DTC Performances prediction	Updated DTC	Standard Compliance review Product Configuration and option
Target margin for the unit Preliminary datasheet or Project Goal Sheet	List of applicable standard List of deviation to the original target if any	Test plan	List of modification	Sales documentation Unit documentation